FINAL PROJECT REPORT

SUPERSTORE SALES DATA ANALYSIS

ApexPlanet Data Analytics Internship

Project: Superstore Sales Data Analysis

Tools: Python, Pandas, Matplotlib, Seaborn, SQL, Tableau, Scikit-learn

Dataset: Sample Superstore

Report scope: Tasks 1–4

Prepared as the final reporting deliverable for the data analytics project.

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1. Executive Summary

This project analyses the Sample Superstore dataset to understand sales, profitability, customer behaviour and business performance. The work covers data preparation, exploratory data analysis, SQL-based analysis, visualisation, statistical testing, time-series forecasting, customer segmentation and predictive modelling.

The analysis found substantial variation in sales and profit, a significant difference in profit between discounted and non-discounted transactions, and useful customer groups with different purchasing and profitability patterns. Among the tested forecasting models, Exponential Smoothing produced the lowest holdout errors.

The results can support better discount decisions, customer retention, product planning and sales monitoring. The findings describe patterns in the available dataset and should not automatically be interpreted as causal relationships.

2. Problem Statement and Objectives

Problem statement: Understand the factors associated with Superstore sales and profitability and identify useful patterns that can support business decisions.

Objectives:

• Prepare and clean the dataset for analysis.

• Explore sales, profit, quantity, discount and customer patterns.

• Use SQL queries to retrieve and analyse business information.

• Create visualisations and an interactive dashboard.

• Apply statistical tests and time-series forecasting.

• Segment customers and evaluate predictive machine-learning models.

• Present actionable recommendations based on the findings.

3. Dataset and Methodology

The project uses the Sample Superstore dataset. The raw Excel file is stored in data/raw/sample_-_superstore.xls and the cleaned dataset is stored in data/processed/superstore_cleaned.csv.

The workflow includes data inspection, cleaning, exploratory analysis, SQL querying, visualisation, statistical testing, forecasting, segmentation and model evaluation.

4. Data Cleaning and Exploratory Analysis

The preparation work included standardising column names, checking missing values and duplicates, reviewing data types and dates, and examining outliers using the IQR method. Legitimate outlier observations were retained.

The exploratory analysis examined sales distribution, category and regional performance, profit patterns, customer segments, sales versus profit and correlations.

Recorded EDA observations: Sales are strongly right-skewed. Technology records the highest total sales and profit among categories. The West region and Consumer segment record the highest total sales and profit in the initial analysis.

5. SQL Analysis and Visualisation

Task 2 used SQL queries and Python integration for structured data analysis. The SQL query file is available at scripts/task2_queries.sql. Database helper scripts are stored in scripts/.

Task 3 developed visualisations and a Tableau dashboard. The project contains dashboards/Task3_Final_Dashboard_Backup.twbx and the monthly sales trend HTML visualisation. The dashboard should be opened in Tableau to review its interactive elements.

6. Statistical Analysis

The Task 4 analysis calculated descriptive statistics and confidence intervals and conducted hypothesis tests.

Measure	Mean	Median	Standard deviation
Sales	228.226	53.91	619.907
Quantity	3.792	3	2.228
Discount	0.155	0.20	0.206
Profit	28.673	8.69	232.465

The 95% confidence interval for mean Sales was 216.19 to 240.26.

Test	Result	Interpretation
Discounted vs non-discounted profit	t = 15.8941; p = 3.73 × 10■■■	Statistically significant difference
Segment vs Region	χ^2 = 6.1988; p = 0.4013	No significant association detected
Profit across Segments	F = 0.9414; p = 0.3901	No significant mean difference detected

Average profit was 66.34 for non-discounted transactions and -6.53 for discounted transactions. This is an observed association and does not establish that discounts alone caused the difference.

7. Time-Series Forecasting

The monthly sales series contained 48 observations from January 2023 through December 2026, with total sales of 2,326,534.35. The Augmented Dickey-Fuller test returned a statistic of -4.4375 and p-value 0.000255.

Model	MAE	RMSE	MAPE
Exponential Smoothing	11,752.42	13,071.68	22.52%

Model	MAE	RMSE	MAPE
ARIMA	23,717.82	27,320.37	53.96%
SARIMA	13,642.80	15,996.53	26.61%

Exponential Smoothing had the lowest holdout errors among the three tested models. Its six-month forecast for January–June 2027 totals approximately 372,170. These are model forecasts, not guaranteed sales.

8. Customer Segmentation

K-Means clustering was applied to 804 customers using total sales, total profit, total quantity, average discount and number of orders. Features were standardised before clustering. Four clusters were selected for business interpretability, although k=2 had the highest silhouette score among the tested values.

Cluster	Customers	Avg. Sales	Avg. Profit	Interpretation
0	202	1,551.19	-96.28	Lower sales, loss-making
1	264	3,712.55	367.00	Higher volume, profitable
2	55	9,717.80	2,281.43	High-value customers
3	283	1,761.84	315.83	Lower sales, profitable

The first two PCA components explained 73.36% of the variance. The cluster profiles suggest different retention, cross-selling and discount-review strategies.

9. Machine Learning Results

Model	MAE	RMSE	R^2
Linear Regression	61.709	214.026	0.489
Decision Tree Regressor	28.787	120.929	0.837

The Decision Tree Regressor performed better than Linear Regression on the reported test metrics. The leading feature importances were Sales (0.604) and Discount (0.275). Feature importance indicates model reliance, not causation.

The Logistic Regression model classified transactions as above or below the median profit threshold of 8.69. It achieved accuracy 0.797, precision 0.7911, recall 0.8067, F1-score 0.7988 and ROC-AUC 0.8833. This is a high-profit classification task, not churn prediction.

10. Key Findings and Business Recommendations

• **Review discount practices:** Discounted transactions had lower average profit in this dataset. Evaluate discount levels alongside product costs and margins.

• **Retain high-value customers:** The high-value cluster had the highest average sales and profit. Consider loyalty and retention initiatives.

• **Improve loss-making customer economics:** Review discounts, costs and order composition for the loss-making cluster.

• **Grow profitable lower-sales customers:** Use relevant cross-selling and bundles to increase order value.

• **Use forecasts carefully:** Exponential Smoothing was the best of the three tested models on the reported holdout metrics. Monitor forecast errors as new data arrives.

• **Monitor performance:** Use the dashboard to track sales, profit, regions, categories and customer segments.

11. Limitations and Future Scope

The analysis is limited to the fields, observations and time period available in the supplied dataset. Statistical associations do not prove causation. Forecasts depend on historical patterns and may change when business conditions change. Clustering results depend on feature selection and the chosen number of clusters.

Future work could include more recent data, external business factors, additional model validation, automated dashboard refreshes, monitoring of forecast performance and deployment of the analytics workflow.

12. Conclusion

The project demonstrates an end-to-end analytics workflow using data preparation, exploratory analysis, SQL, visualisation, statistical methods, forecasting, clustering and machine learning. The findings provide a basis for reviewing discount strategies, understanding customer groups and monitoring sales performance.

Appendix: Project Visualisations

The following figures are included from the existing project chart exports. Refer to the relevant notebooks for each chart's exact analysis and interpretation.

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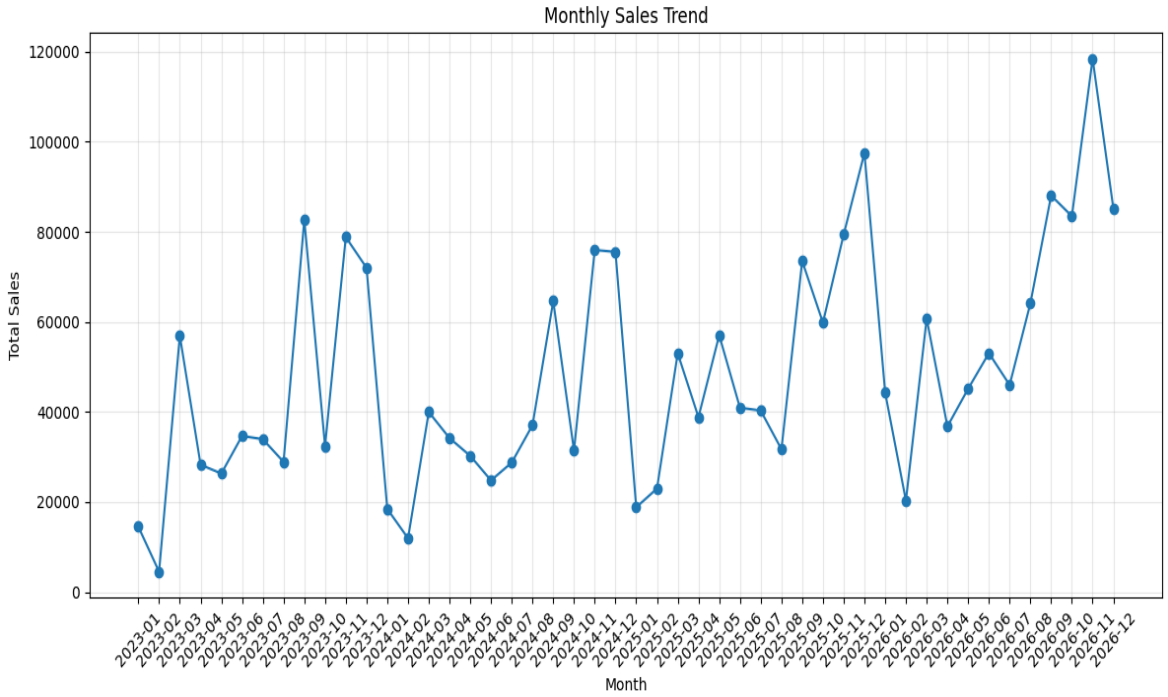


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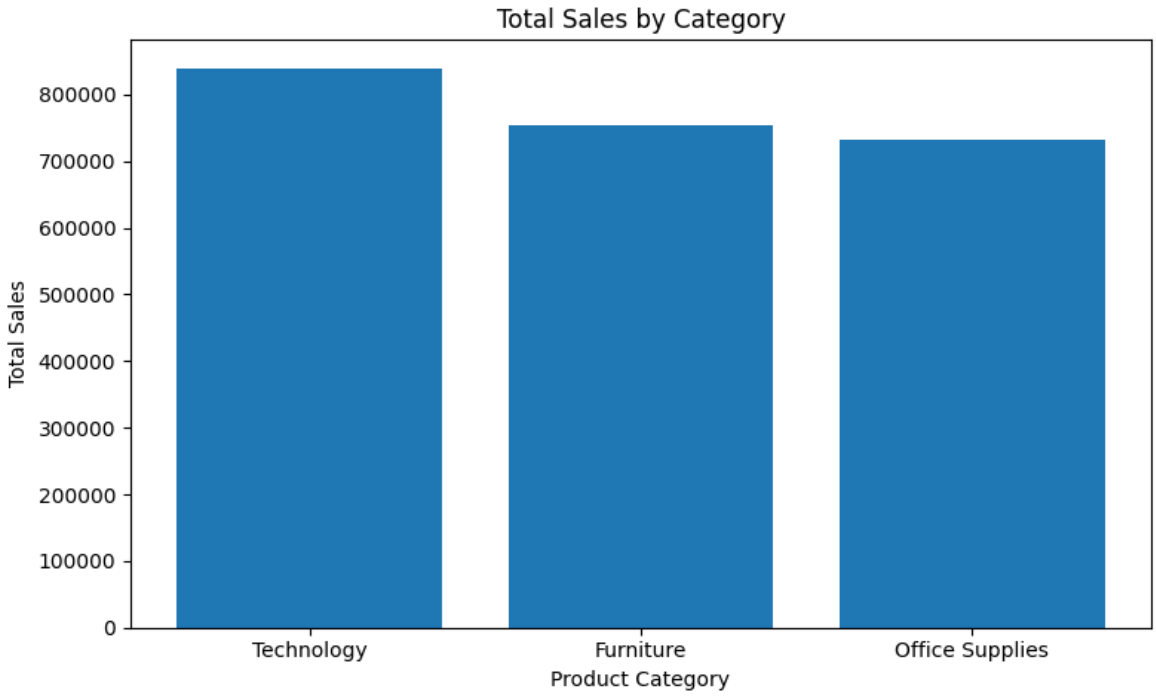


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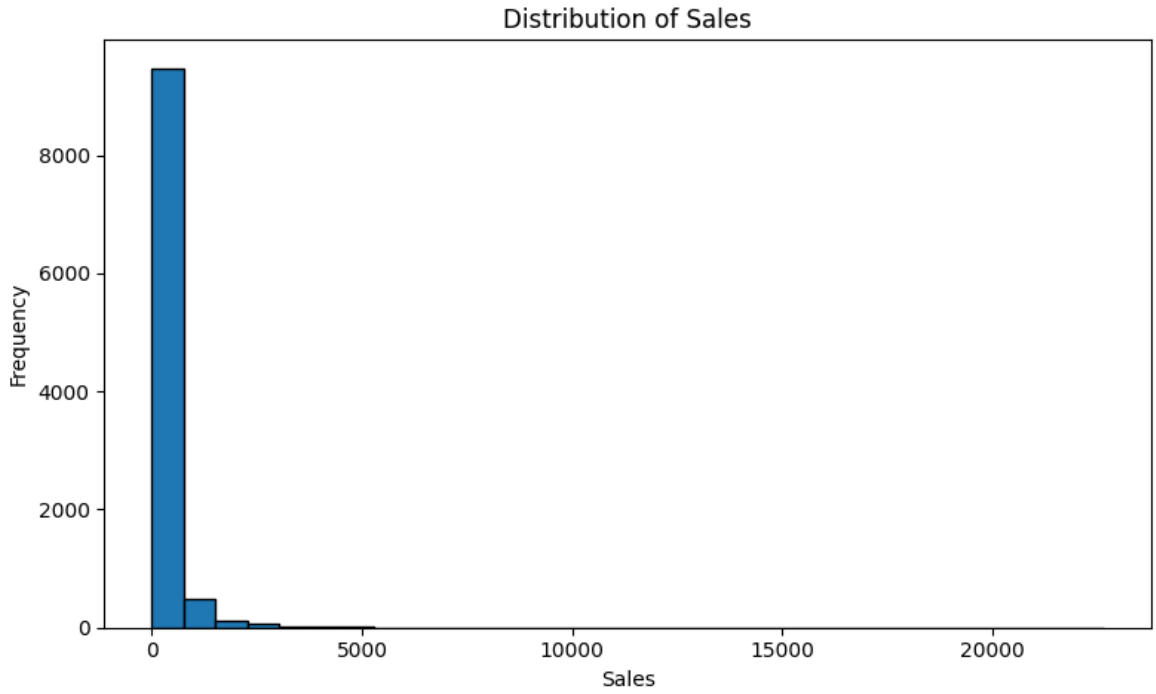


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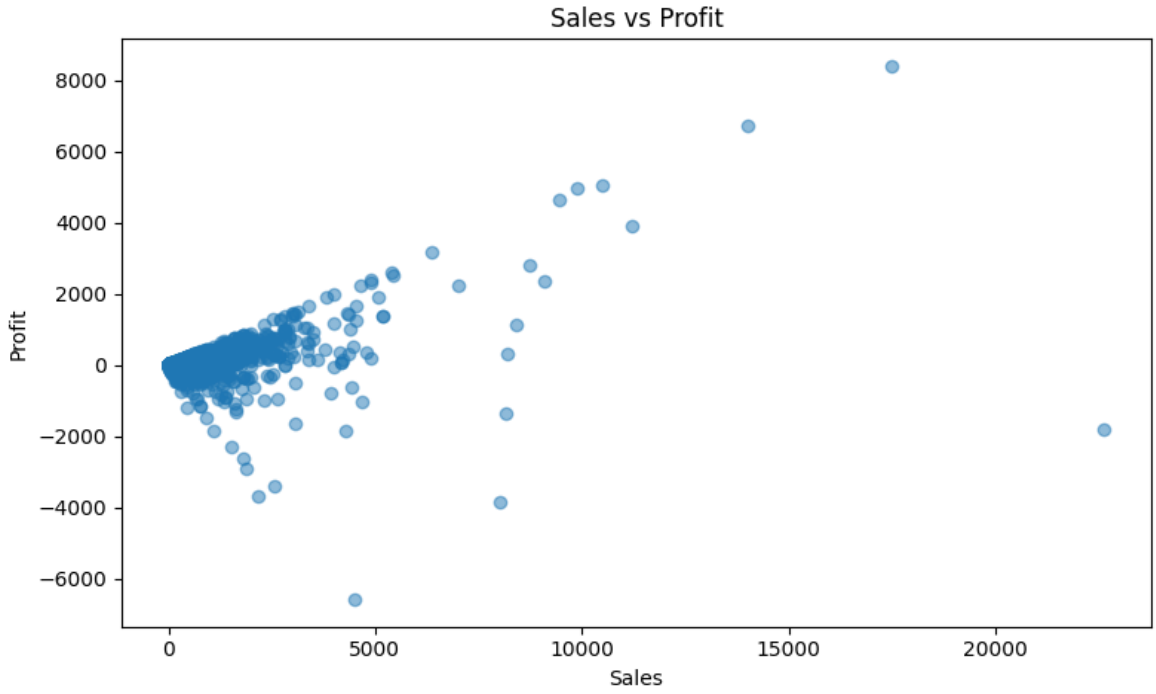


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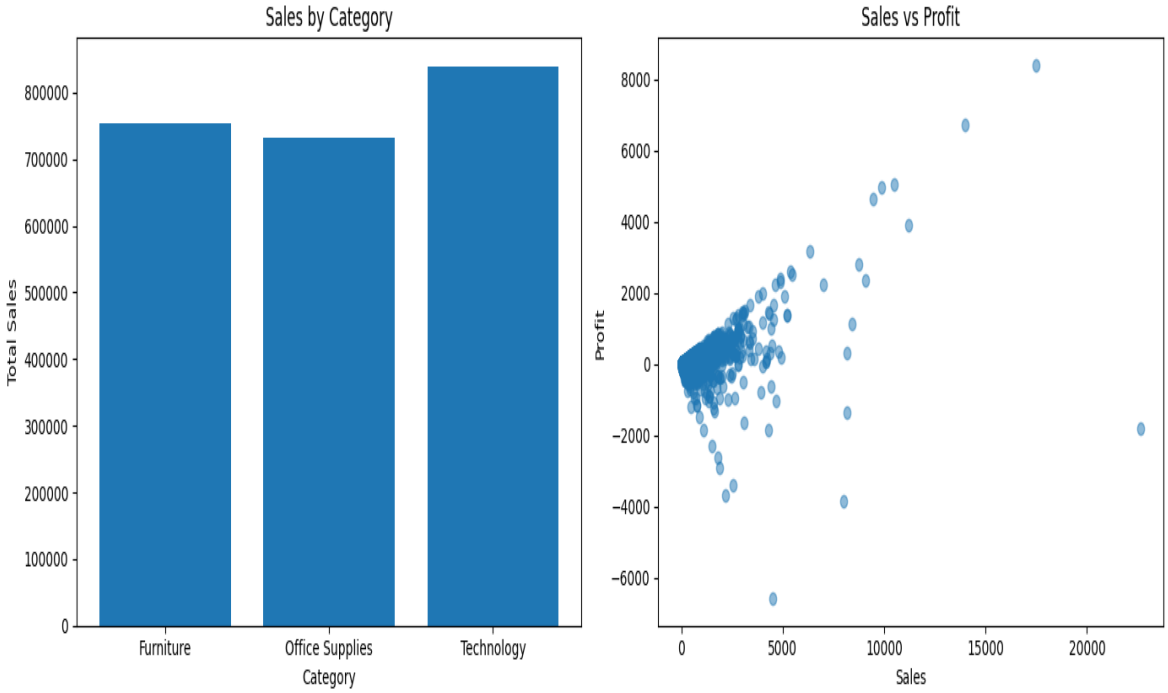


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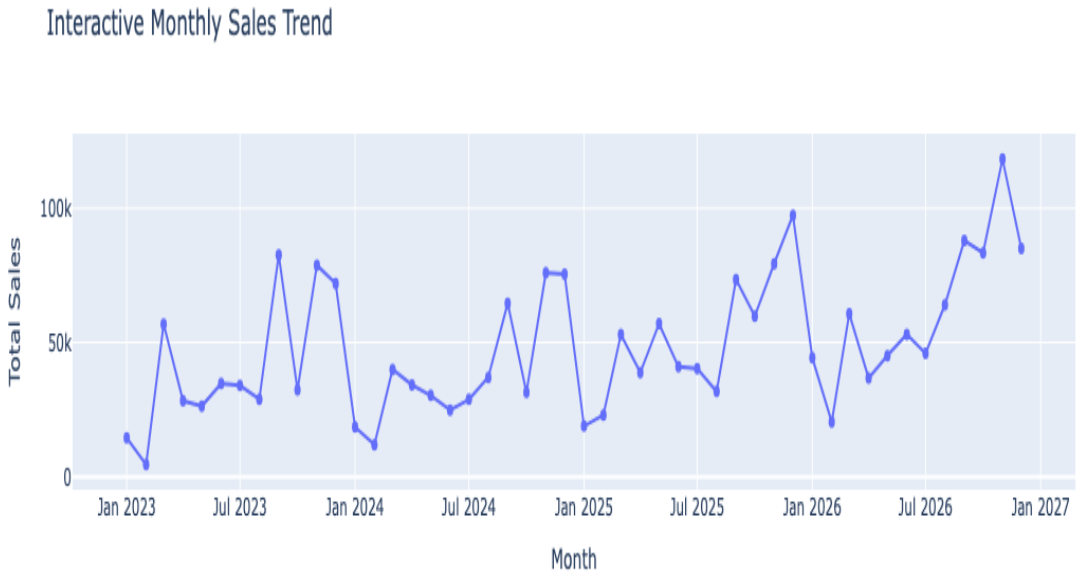


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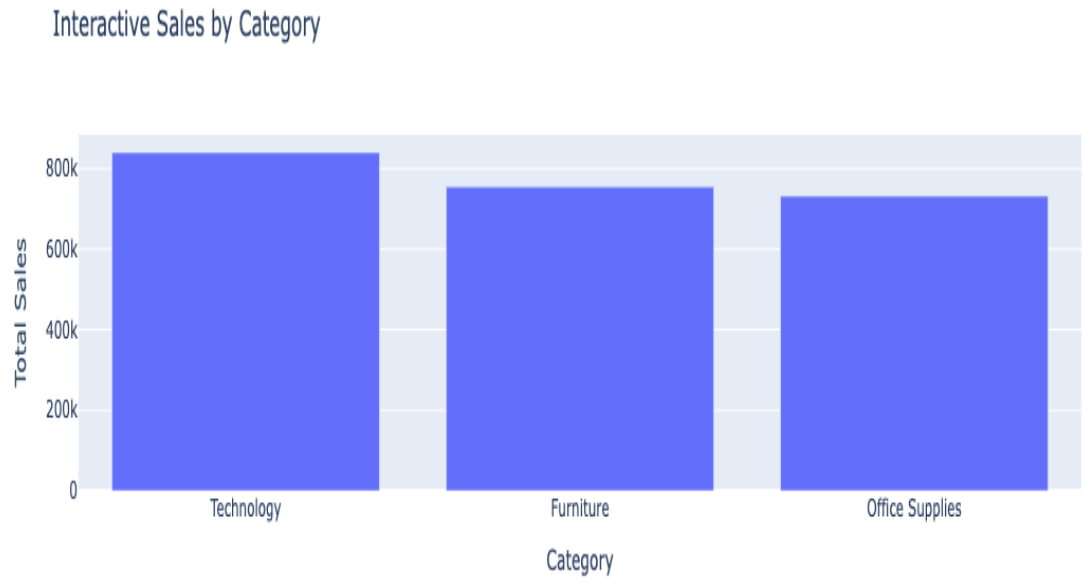


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chart_09.png

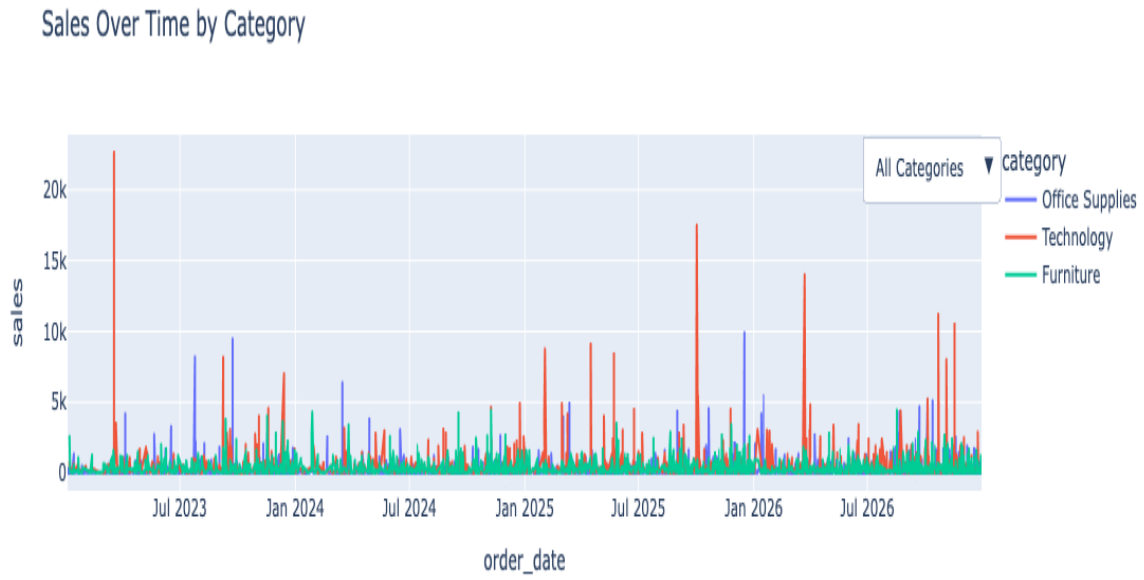


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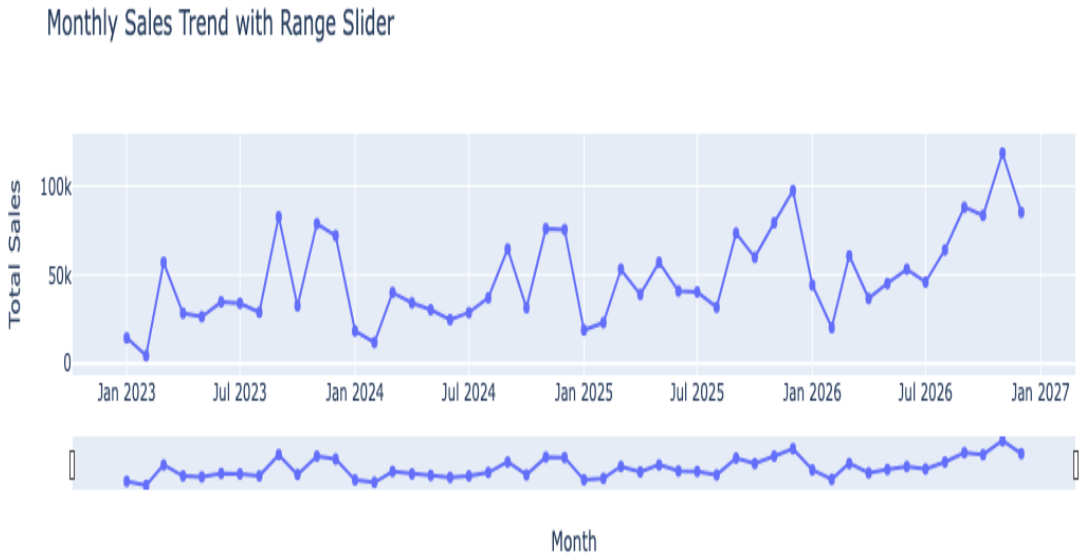


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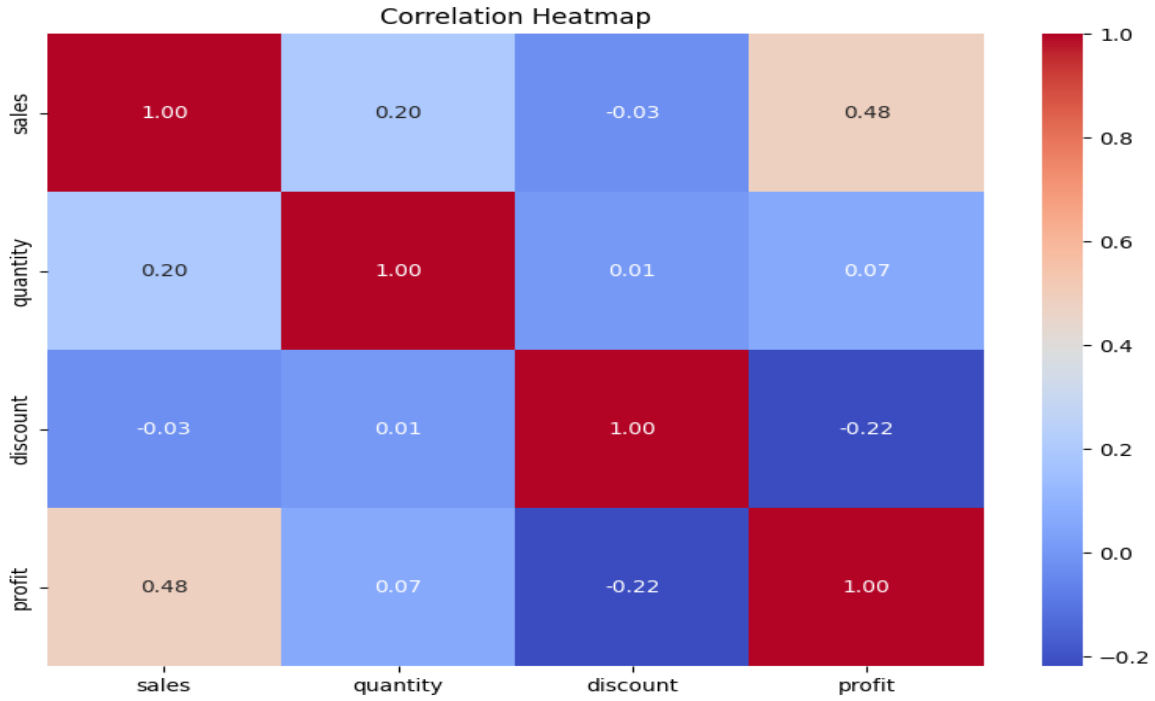


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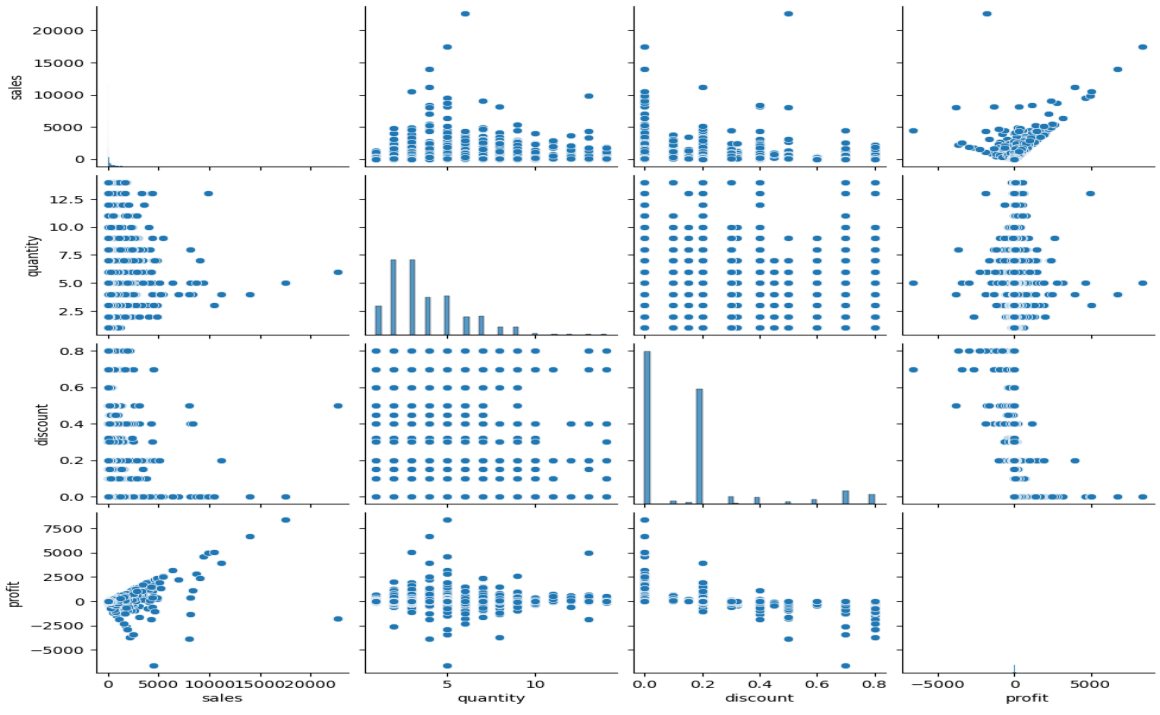


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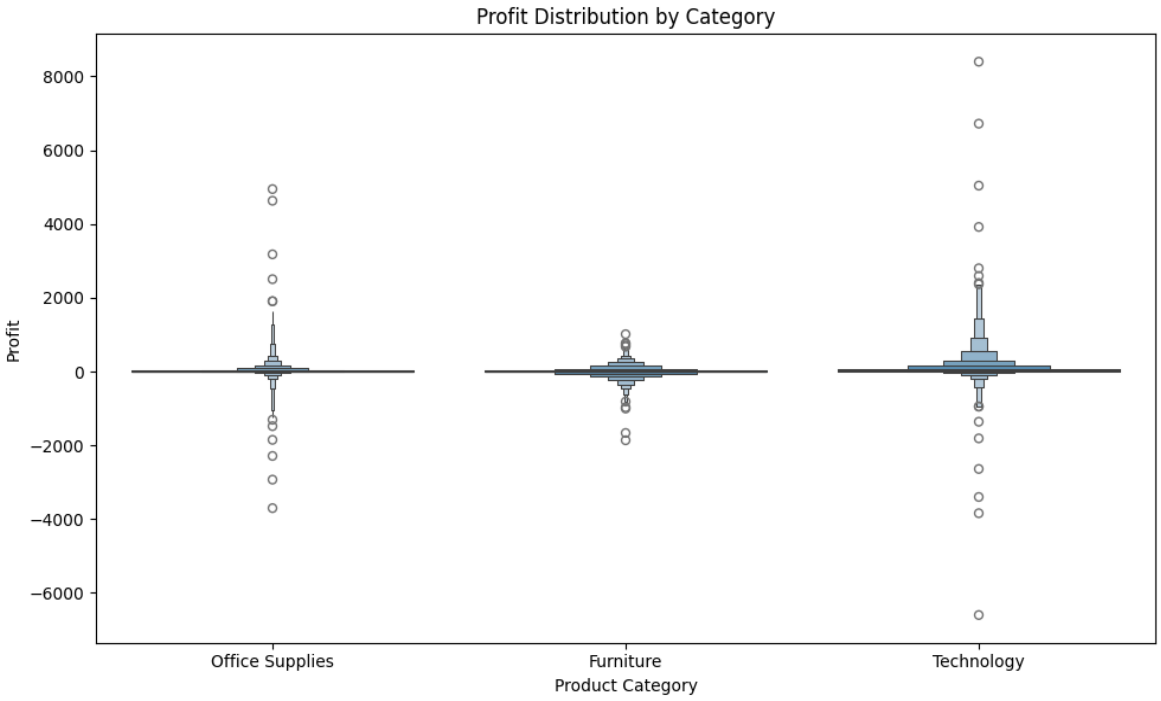


chart_14.png

